Rihal OutSystems Challenge – Leave Management System

Project Overview

The Leave Management System is a web application built using OutSystems that allows employees to request leave and administrators to manage approvals efficiently. The system automates leave tracking, reduces manual paperwork, and provides real-time visibility of leave balances.

System Access / Demo Credentials

Role	Username	Password	Description
Admin	Admin	Admin123	Manage leave types, monitor all requests, dashboards.
Manager	Manager	Manager123	Approve or reject employees' requests.
Employee	Employee	Employee12345678	Create and manage and track own leave requests.
Employee	Employee2	Employee2123	Create and manage own leave requests.

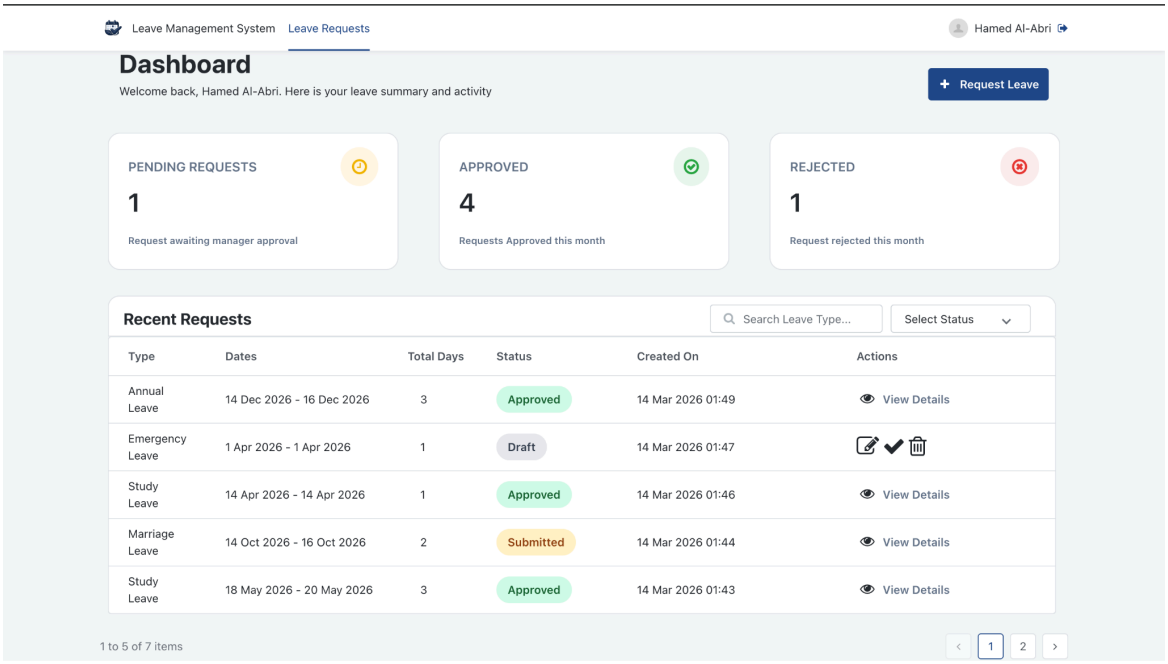
Application URL:
https://personal-vakhoryl.outsystemscloud.com/LeaveManagementSystem/Login

Objectives

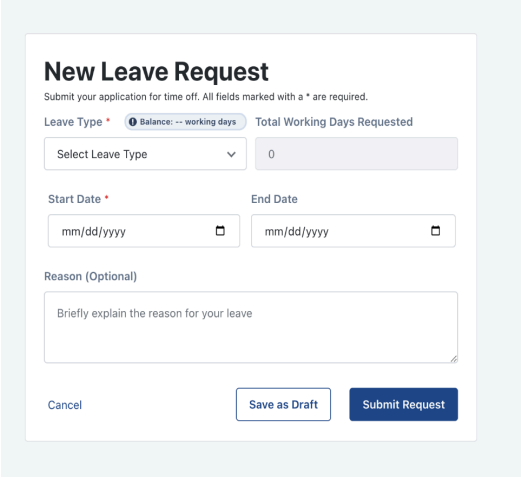
- Automate employee requests
- Provide real-time leave balance tracking
- Simplify approval workflows
- Reduce administrative workload

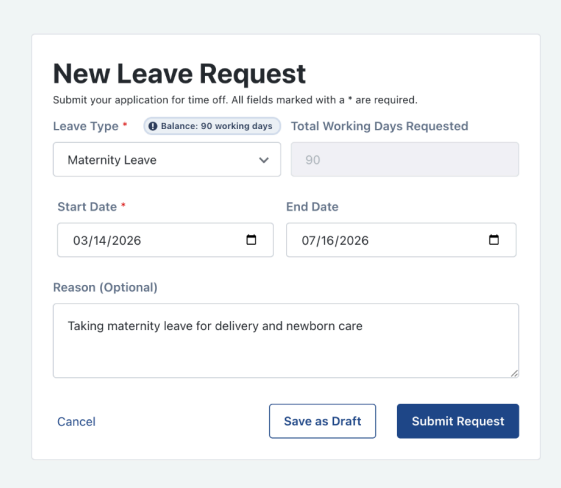
System Features

Employee features



An employee can save or submit a leave request, track their request status, and view their request/leave history. If the employee chose to save their request it will automatically be in the draft staus and the user can edit their request, submit it or delete it. Edits can only happen while in the draft status. Once a user submits their request they will only be able to view the details of their request and track the status. A small dashboard shows the pending requests, requests approved this month and requests rejected this month. The user can also use the search bar in the table to search for records or filter based on status

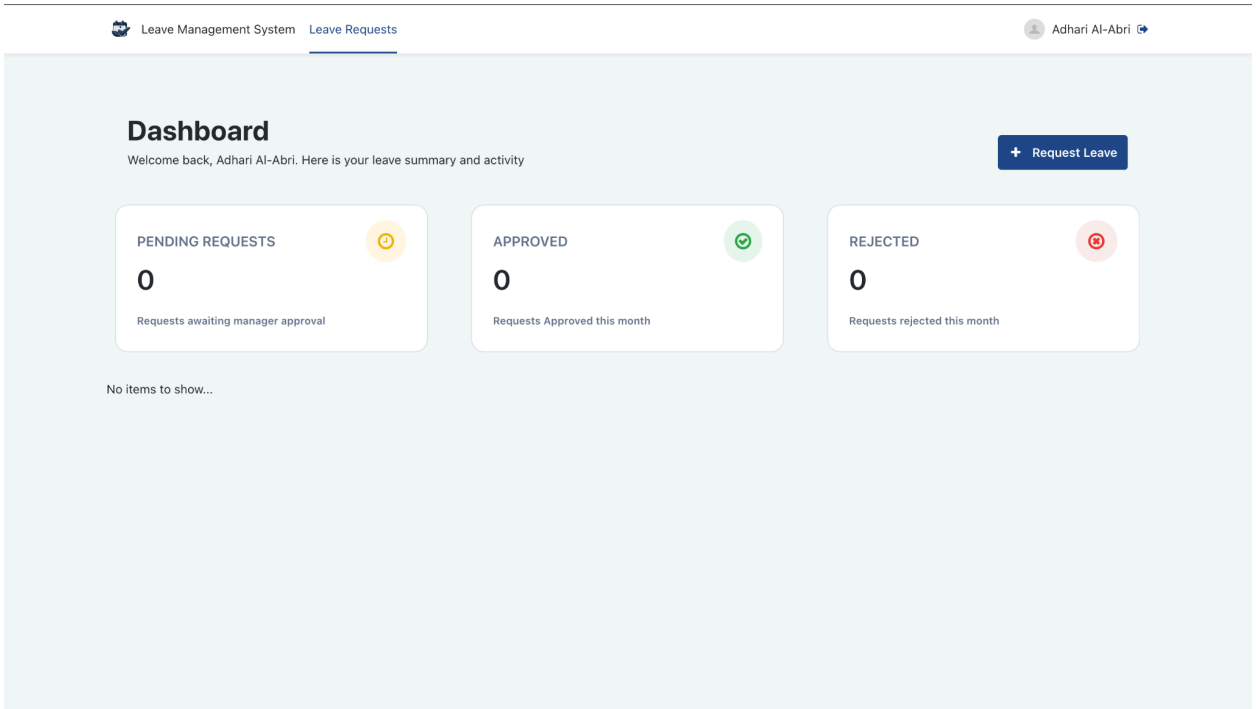




The form in which the user submits or saves their request has the following features

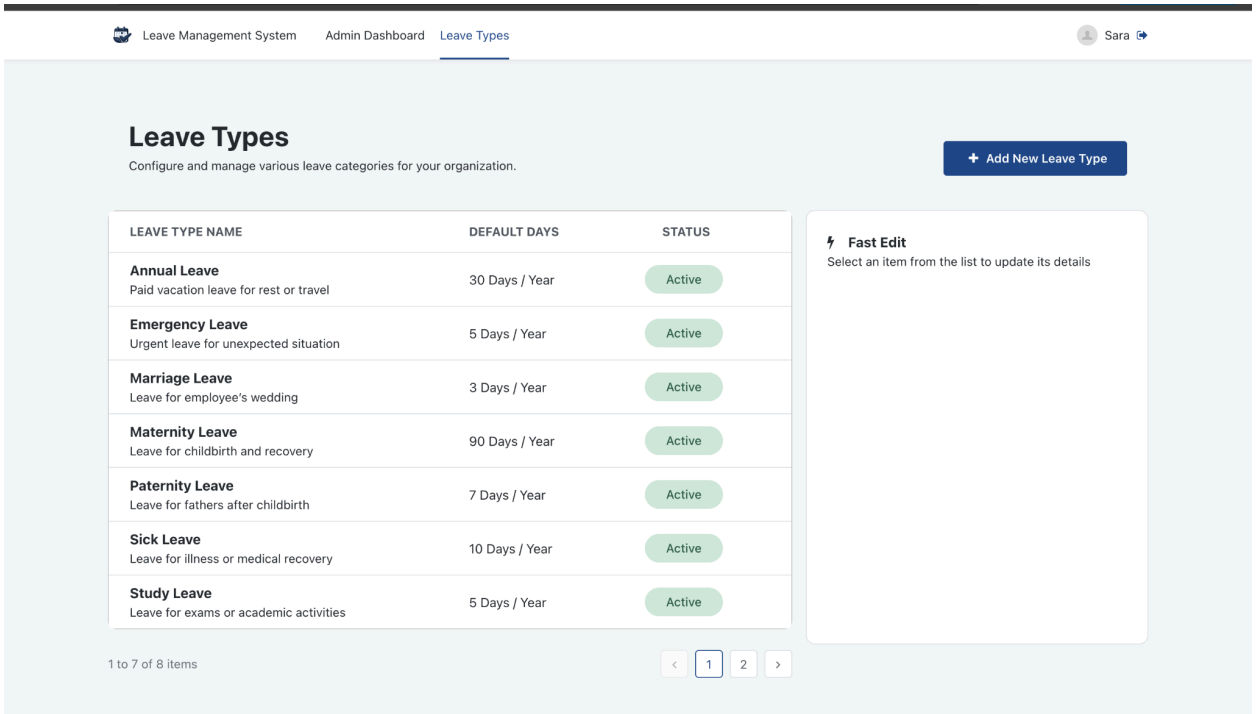
- **Balance calculation** based on the leave type selected
    - If the leave type has not been used before, then the balance is the default days of that particular leave type set by the admin

    - If the leave type has been used before then the balance is calculated as the default days per year set for that leave type minus the approved working days from the previous requests using that leave type
- **Automatic calculation of the total working days** - once the start date and end date have been set, then the total working days will be shown. The working days is calculated using the start and end date and **excluding the weekend (friday and saturday)**
- If the user's balance for that particular leave type is 0 then he will not be allowed to save or submit his request and will be prompted to change the leave type chosen. And if the number of total working days exceeds the balance then the user is prompted to adjust the selected dates to reduce the number of working days. These validations ensure that all leave requests comply with the user's available leave balance.
- Another validation was to ensure that the starting date chosen has to be before the end date, and vice versa. Additionally, the starting date chosen cannot be a date before today, ensuring that managers have enough time to accept or reject the request.
- When the employee selects a start date, the end date is automatically set to the same date. The user can then modify the end date if a longer leave period is required. This ensures that the minimum leave duration is one working day.
- The drop down input of the leave type only shows active leave types

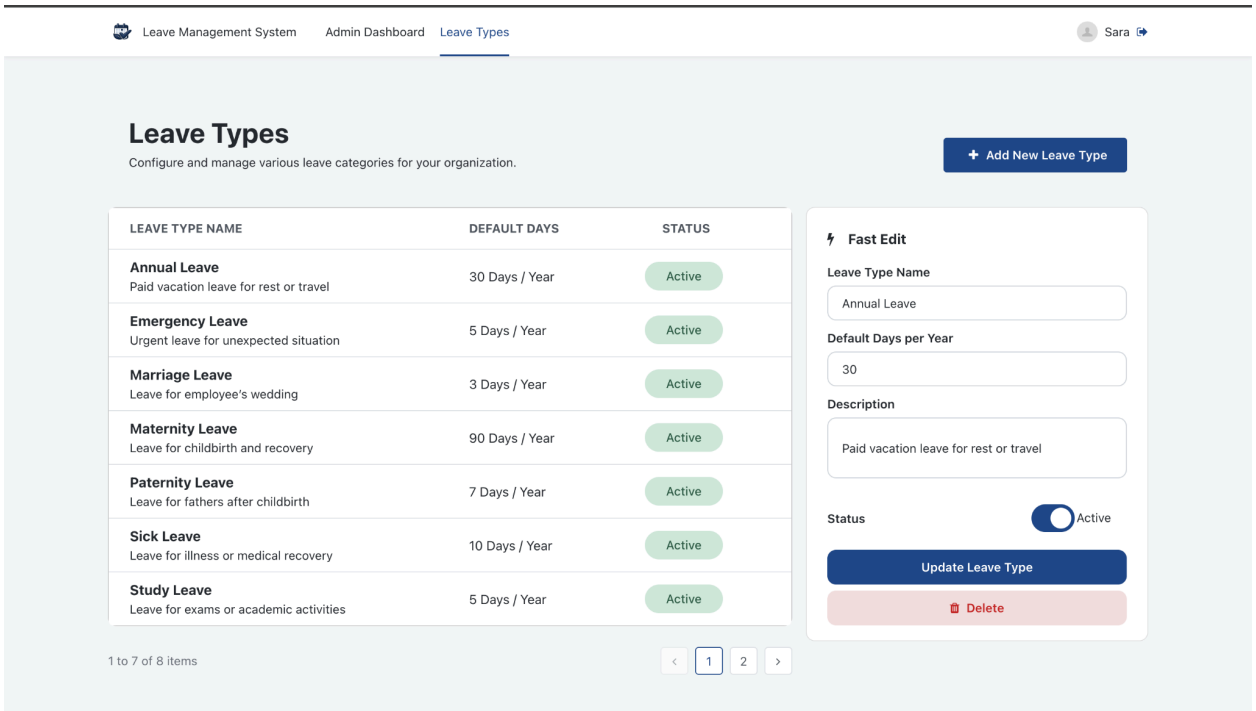


If the user has no requests, then this is how the dashboard will look like

Admin Features

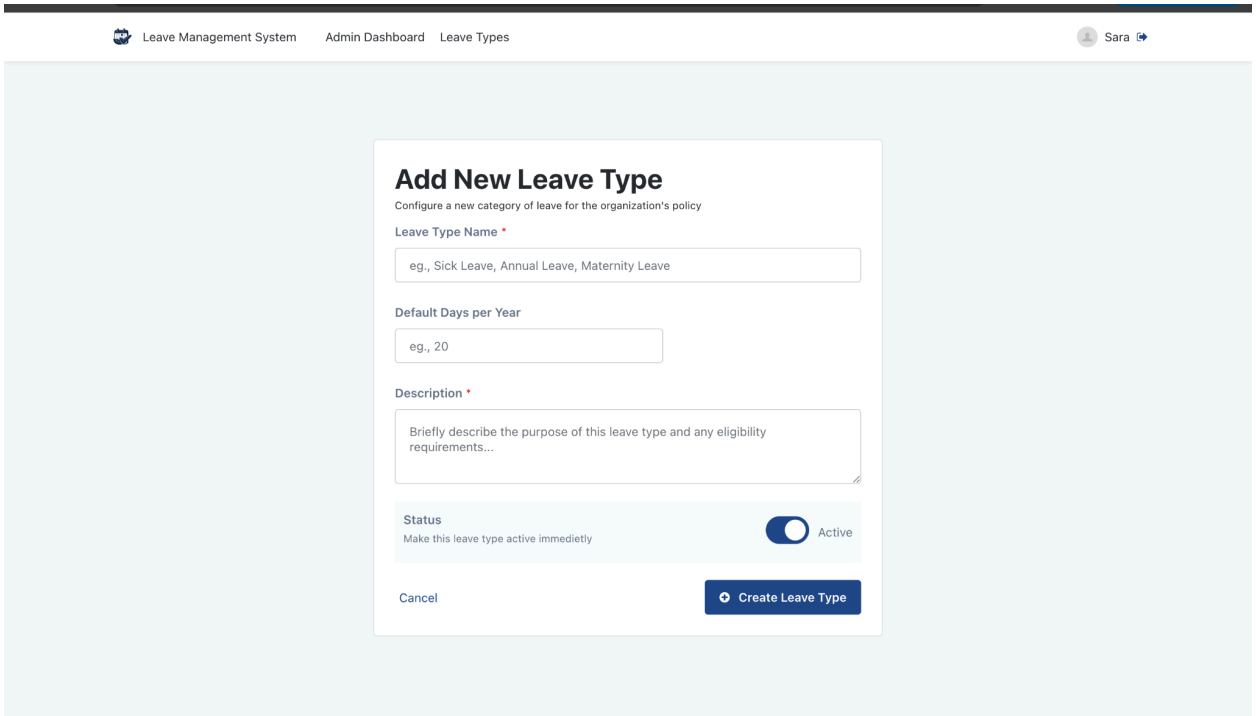


The admin can manage leave types and view the name, description, default days and status. There is also a fast edit feature where the admin can click on one of the leave type rows and edit its details as shown below

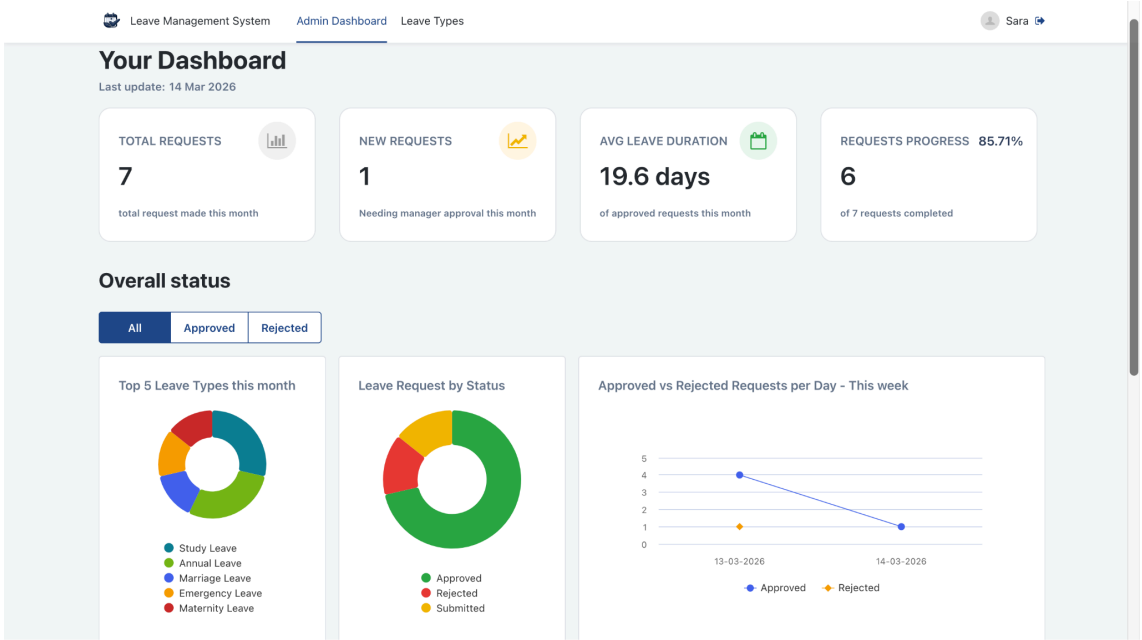


The admin will be able to update the leave type or delete it. If the admin deactivates a leave type then it will be hidden from the drop down of leave types in new user requests. Also, the

delete button uses a soft delete feature so that the history of previous requests using that leave type is still maintained.

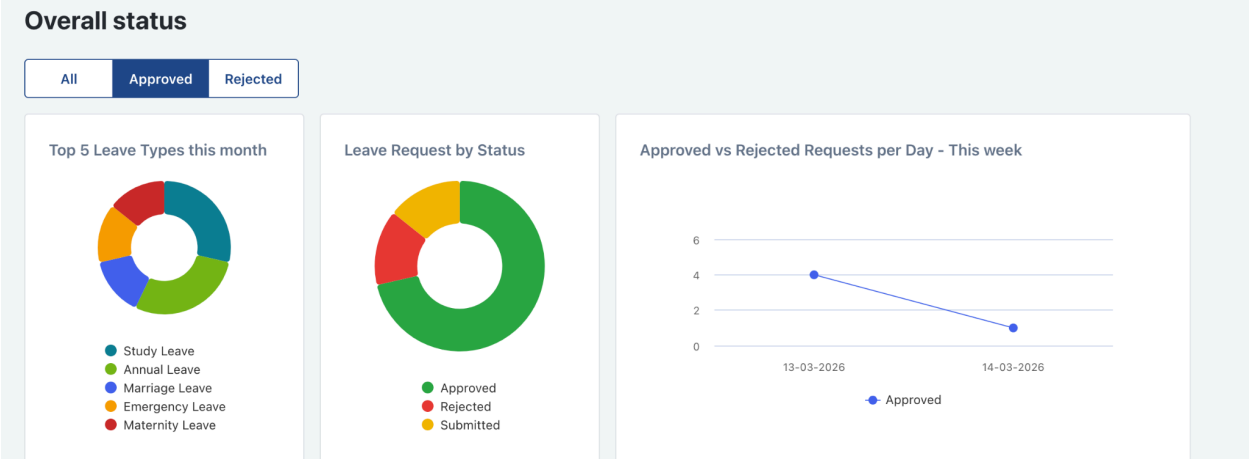


Upon clicking on the add new leave type button, the admin will be redirected to this form where they will be prompted to add the name, default days per year, description and status.

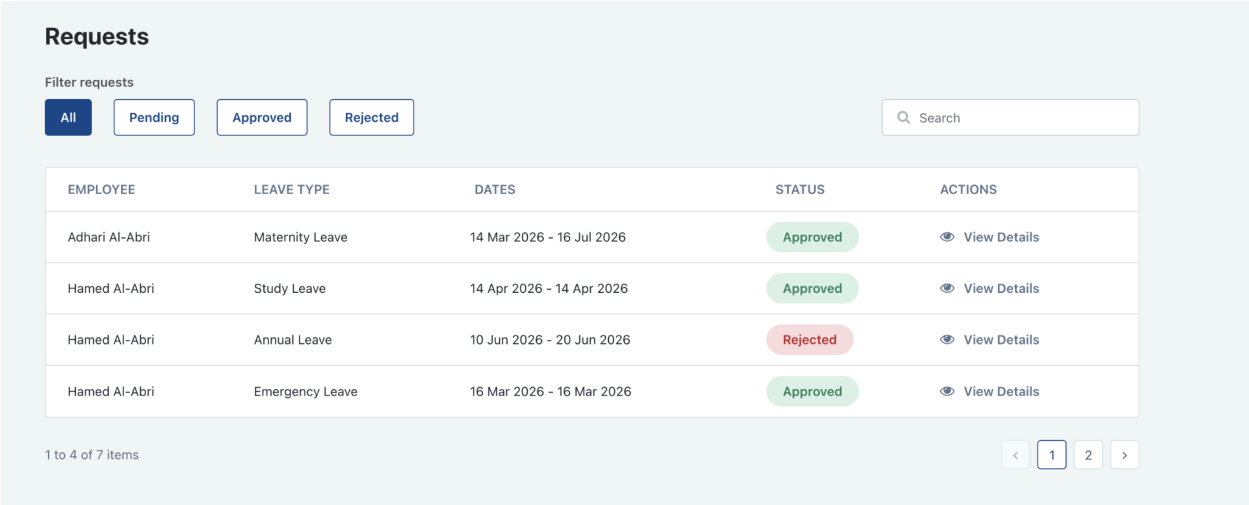


The admin will also have a general overview of leave stats using this dashboard. The first part of the dashboard shows total requests made this month, new requests this month requiring manager approval and rejection and the request progress so that the admin can track the

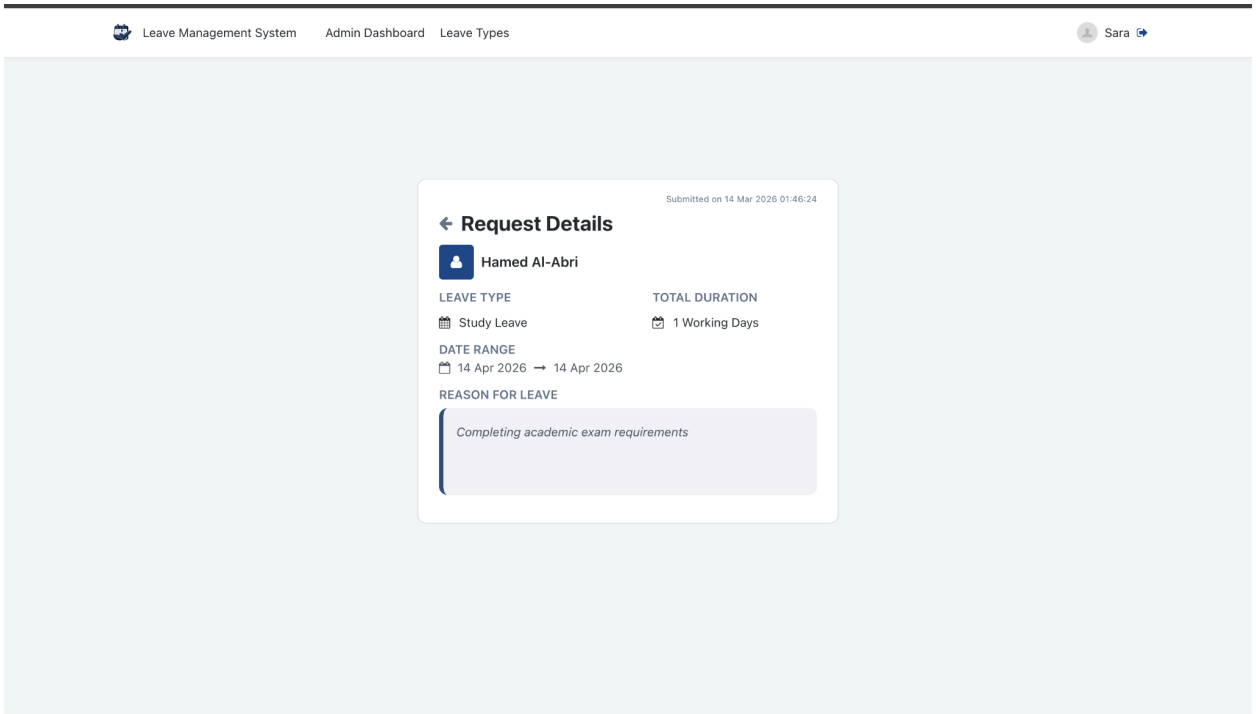
managers decisions and progress. It also shows the average leave duration of approved requests this month.
The second part of the dashboard shows statistical graphs. The first chart shows the top 5 leave types requested this month and their distribution based on submitted, approved and rejected requests. The second chart shows the leave request by status using the current month's data and the third line graph visualises the approved vs rejected requests per day for the current week. The button group also changes the line graph accordingly to show the numbers in more detail as shown in the picture below



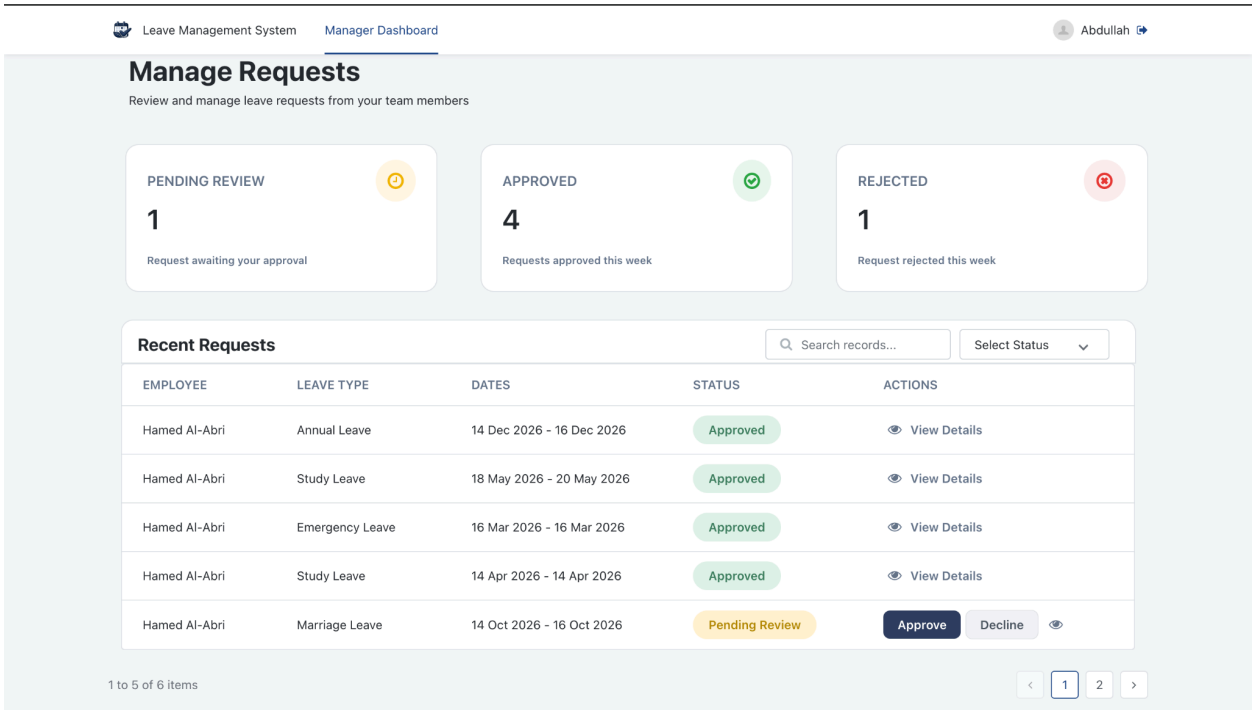
Finally, the last part of the dashboard shows all requests made by various employees so that the admin can track these requests and can filter then by status or search for a particular record.



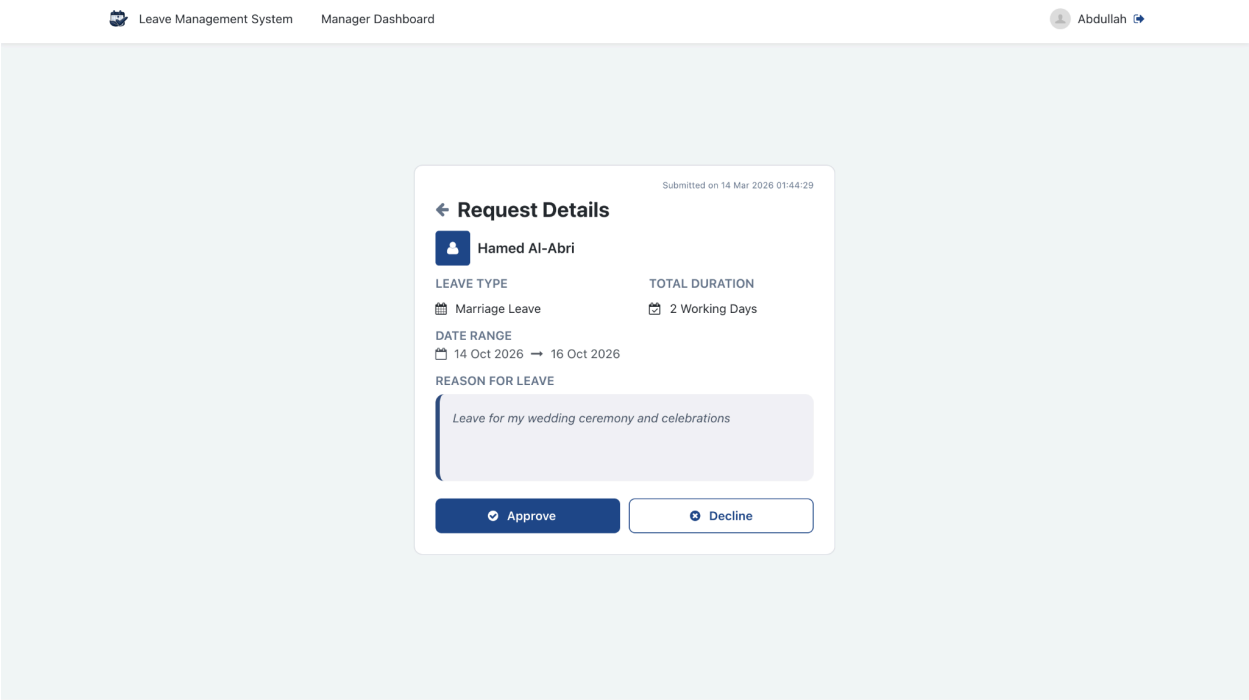
The admin can also click on the view details button where he will be redirected to the details page, where it looks the same as the details page in the admin dashboard but without the approve or reject buttons as those are particular to managers only



Manager features



The manager dashboard will help the manager track requests by showing requests that are pending approval, approved requests this week and finally, the rejected requests this week. The manager will be able to search records and filter by status, approve or decline request on the spot. Or click on the eye icon to view details and then approve/decline accordingly as shown in the screen below



Challenges and solutions

Challenge 1: dynamic leave balance calculation ensuring that the system accurately calculates the remaining leave balance for each employee, especially when multiple leave requests have already been approved.

Solution: A client side action was implemented that retrieves the default leave days for the selected leave type and subtracts the total number of approved leave days for that employee. This ensures the leave balance is updated dynamically whenever a new request is selected from the drop down

Challenge 2: handling null values in leave calculations - When an employee selects a leave type that has not been used before, the system initially returned a null value for the approved leave days, which caused incorrect balance calculations.

Solution: A validation condition was added to check if the count of approved leave days is null. If no records exist, the system defaults the value to zero before performing the calculation.

Challenge 3: maintaining historical data for inactive leave types - When editing an existing leave request, some leave types may have been marked as inactive in the system. Since inactive leave types are normally hidden from selection lists, this created a problem where previously saved leave requests could not correctly display their original leave type when opened in the edit screen.

Solution: To address this issue, the system was designed to still retrieve and display the leave type associated with the existing leave request, even if the leave type is currently marked as

inactive. While inactive leave types are excluded from the dropdown for new selections, the previously selected value is preserved and displayed to maintain historical accuracy and data integrity.

Future improvements

Future improvements could include email notifications, multi-level approval (if needed), department-based management approval, and carry-forward of unused leave days for this year's leave type.

Conclusion

This was a very fun project to work on, and it was definitely a learning curve for me as I only got recently introduced to the OutSystems platform. As a third year student myself, it was really nice to apply the theoretical knowledge gained in uni into practice and gain practical experience with low code development. I think the biggest challenge was trying to balance my uni work and completing this challenge on time within the given timeline but overall I am very proud of this project and the progress I've made in these 2 weeks, and if you are still reading this then I hope you have a wonderful day!

Thank you Rihal for this challenge and I will definitely be participating in the next codestacker challenge!